

Noah James

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Education

Bachelor of Science, Software Engineering; Minor in Entrepreneurial Studies

May 2028

Rose-Hulman Institute of Technology, Terre Haute, IN

GPA: 3.35

Honors: Dean's List (Spring 2025, Fall 2025, Winter 2025-2026, Spring 2026)

Relevant Courses: Data Structures & Algorithm Analysis, Software Quality Assurance, Software Requirements Engineering, Software Project Management, Operating Systems, Intro to Web Programming, Practical Software Maintenance, Database Systems, Computer Architecture I, Object-Oriented Software Development

Skills

Languages: C, CSS, HTML, Java, JavaScript, Python, SQL, TypeScript

Software: Git, React, Visual Studio Code, Eclipse, Linux, OnShape, AutoCAD, Google Workspace, Test-Driven Development

Project Experience

PitchVision - <https://github.com/rhit-jamesnc/PitchVision>

August 2026-Present

- Built an automated computer vision framework in Python using YOLO and ByteTrack to extract frame-by-frame player and ball tracking data from soccer footage.
- Applied homography matrices to project 2D camera pixel coordinates onto a top-down coordinate grid, measuring real-world pitch metrics in meters.
- Engineered predictive decision models (Logistic Regression, Expected Pass Completion, Decision Quality Score) using spatial vector math and Voronoi cell area analysis.
- Developed event segmentation logic to capture frame T0 (ball receipt) and evaluate passing, shooting, and dribbling options at the moment of possession.
- Followed a Test-Driven Development (TDD) workflow to validate coordinate mapping transformations, passing lane geometries, and decision engine analytics.

TopDown Tactix - <https://topdown-tactix.vercel.app/>

June 2026

- Architected a real-time, 2D multiplayer soccer engine using a monorepo architecture with React (frontend) and Node.js/Socket.io (backend) to achieve high-fidelity client-server state synchronization.
- Engineered a high-performance, deterministic physics engine in Matter.js, implementing sub-stepping, custom collision filtering, and velocity clamping to ensure simulation stability during high-speed interactions.
- Developed a difficulty-scaled AI controller featuring reactive logic and cognitive delay modeling; implemented behavior patterns that dynamically adjust to player performance for balanced gameplay.
- Established a comprehensive testing suite using Vitest to validate critical game systems, including edge-case goal detection, state resets, and server-side socket communication.
- Managed the end-to-end production lifecycle, including automated CI/CD pipeline integration via Vercel/Render, Vite bundling, and Git-based kanban workflows for iterative feature development.

Forbidden Island

March 2026-May 2026

- Engineered a desktop implementation of the board game Forbidden Island in Java, managing complex game logic through polymorphic role abilities and dynamic state transitions.
- Executed a rigorous Test-Driven Development (TDD) workflow, utilizing mutation testing and static analysis to achieve 100% basis path coverage on core business logic.
- Integrated a ResourceBundle-based localization system to support dynamic language switching without modifying core game mechanics or Swing UI components.

RoseShreddedNerds Workout Tracker

December 2025-February 2026

- Developed a full-stack fitness tracking application for Rose-Hulman students using React and Python, enabling users to log workouts, track personal records, and join fitness classes.
- Engineered a custom RESTful API and HTTP server using Flask and pyodbc to manage secure user authentication, session scheduling, and real-time exercise volume tracking.
- Architected a relational database in SQL Server with stored procedures to automate complex fitness calculations, including Estimated 1 Rep Max via the Epley Formula and campus-wide leaderboards.

- Designed a responsive frontend with Bootstrap and React Router, featuring a dynamic dashboard that visualizes weekly consistency streaks and multi-set workout history.

Portfolio Website - <https://noahcjames.github.io/>

September 2024

- Designed and deployed a responsive portfolio website using HTML/CSS/JS and GitHub Pages, utilizing Git for version control and site maintenance.

Fall Jump Video Game - <https://n04h500.itch.io/jump-fall>

September 2023-December 2023

- Independently designed and developed a full video game from scratch using GameMaker Studio and GameMaker Language (GML).
- Learned and applied a new programming language while implementing core gameplay mechanics, debugging issues, and refining user experience.
- Programmed player behavior and game logic for a complete playable level and published the game online for public access.

Work Experience

Grader – CSSE220 Object-Oriented Software Development, Rose-Hulman - Terre Haute, Indiana

September 2026-Present

- Evaluate and debug Object-Oriented Java homework submissions and programming projects for 51 students, enforcing object-oriented design patterns, clean code principles, and software maintainability.
- Provide detailed feedback on student code quality, recursive algorithms, object inheritance, and JUnit test implementation to support learning outcomes.

Athletic/Event Staff, Rose-Hulman SRC - Terre Haute, Indiana

September 2025-Present

- Set up and tore down multiple athletic and campus events, ensuring venues stayed on schedule.
- Hosted visiting teams and officials, providing clear communication and hospitality throughout events.
- Executed on-site responsibilities reliably to ensure athletic and campus events ran efficiently.

Fire Alarm Installation Technician – Summer Intern, Koorsen Fire & Security

June 2026-August 2026

- Installed fire alarm systems and associated devices according to detailed drawings, blueprints, and specifications.
- Ensured proper functionality of systems while demonstrating adherence to all safety regulations and practices.
- Interacted professionally with customers and vendors to communicate system functionality and service requirements.

Brand Representative, Hollister

June 2026-Present

- Maintain store operational standards to ensure an organized and efficient retail environment.
- Collaborate with team members to deliver high-quality customer service in a fast-paced setting.

Team Member, Graeter's Ice Cream - Bexley, Ohio

June 2022-September 2025

- Collaborated with team members to deliver efficient, accurate customer service in a fast-paced environment.
- Followed standardized procedures to maintain clean, organized operations and reliable store closing.

Certifications and Honors

AP Scholar with Distinction: Awarded for outstanding academic performance on multiple Advanced Placement exams 2023–2024,

Global Seal of Biliteracy – Working Fluency (Spanish): Awarded for high-level proficiency in all language domains.

OHSBC Academic All-Ohio (Varsity Baseball): Awarded for academic excellence while competing at the varsity level (Bexley HS).

"Our Kinda Guy" Award: Selected by baseball coaching staff for team leadership and character.

Leadership Experience

Public Relations and Activities Chair, Interfraternity Council

March 2025-Present

- Directed IFC marketing, event advertising, and social media presence.
- Coordinated IFC philanthropy and campus events in collaboration with the Office of Student Affairs.
- Served as the public voice of the Interfraternity Council, fostering inclusion and campus engagement.

Activities

Rose-Hulman Institute of Technology

- Triangle Fraternity, *Rose Tech Chapter*; Career Services and Employer Relations Volunteer; Soccer Club